Lecture 15 - Optimization I: The First Derivative Test

Garrett Credi

10/29/2026

Updates

Another TA reviews Friday 6–7PM Vincent Hall 16

Grading taking longer than expected, my apologies!
Review Session: 4:30-6:30 in Vincent Hall 16 (the auditorium in the basement of Vincent).
This place is HUGE so feel free to invite your friends in other sections of 1271 to come.
Today's lecture content will ***NOT*** be tested on Exam 2, it is Exam 3 content.

Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz!

Optimization

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4. Comes from *best quadratic approximation*:
$$\mathscr{Q}_{x=x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 \tag{2}$$

curvature = extremes

Towards Optimization

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Given a function $f(x)$, we say that an *input/output pair* $(c, f(c))$ is a ~~*global*~~ local minimum/maximum resp. if,

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Given how similar the defns. are, these two concepts are very tightly related:

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Optimization Defns. Example

Let's differentiate (pun maybe intended) the two defns. by examining the wavy function $f(x) = \frac{\sin(x)}{x}$:

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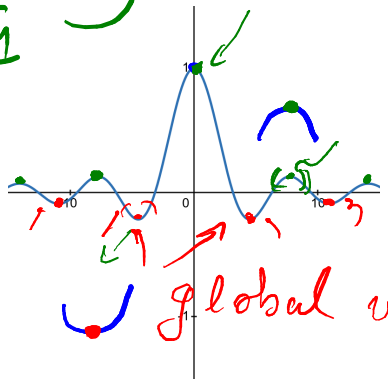
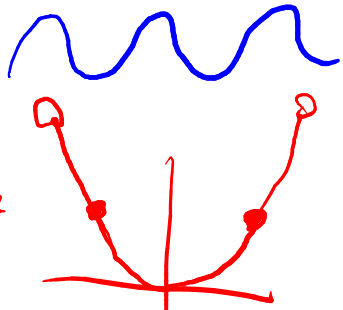


Figure: Graph of $\frac{\sin(x)}{x}$



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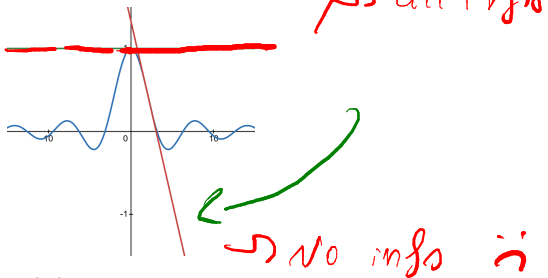


Figure: Graph of $\frac{\sin(x)}{x}$ with two linearizations graphed.

Optimization II

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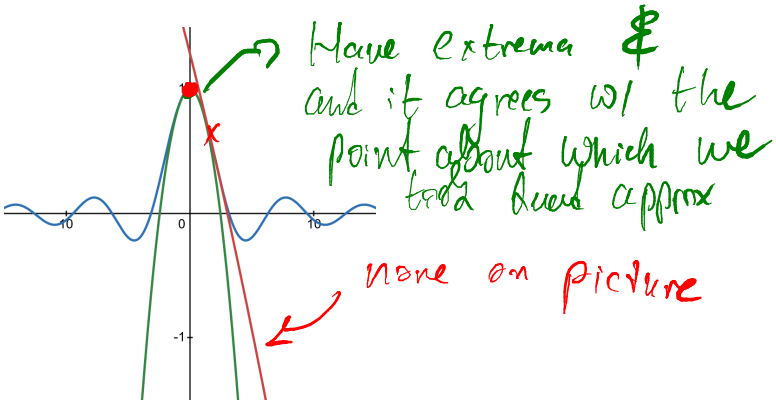


Figure: Graph of $\frac{\sin(x)}{x}$ with two quadratic approximations graphed.

Optimization III

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3. Can we rephrase the condition being a local minimum/maximum in this setting to relate it to conditions on the derivative? Yes!

Optimization IV

Let's assume that $f(x)$ is a function that is both continuously differentiable at $x = x_0$ and that has a local extrema at $(x_0, f(x_0))$.
If we have a local *minimum* at $x = x_0$ then:



$f(x_0) \leq f(x)$

- If $x_0 - \epsilon < x \leq x_0$ (to the left of x_0) then what can we say about $f(x_0)$, $f(x)$ and $f'(x_0)$?

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- If $x_0 \leq x < x_0 + \epsilon$ (to the right of x_0) then what can we say?

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Moral of the story: At a minimum, we start decreasing (negative derivative) then increase (positive derivative) which forces the derivative at the switching point $f'(x_0)$ to be zero.
Exercise: think about what happens when instead $(x_0, f(x_0))$ is a local *maximum*.

Optimization V

We can compile this into the following important result:

First Derivative Test

If $(x_0, f(x_0))$ is a local extremum (or global extremum)

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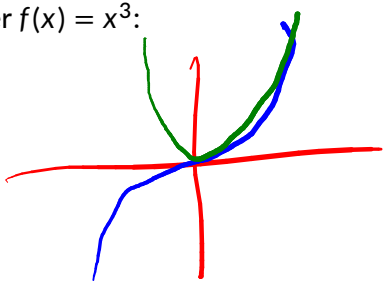
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Unfortunately, this does not *perfectly* classify local extrema, even for the nicest (infinitely differentiable) functions: consider $f(x) = x^3$:

$f'(x) = 0$

$3x^2 = 0$

$x = 0$



Optimization VI

This condition is so important we give it its own name:

Critical Point

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The undefined condition is to handle functions like $f(x) = \sqrt[3]{x^2}$:

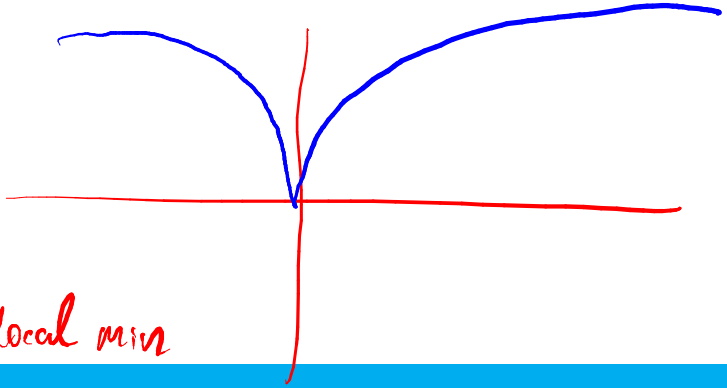
$x^{2/3}$

$f'(x) = \frac{2}{3} \frac{1}{\sqrt[3]{x}}$

$= \text{undf}$

$x = 0$

0 is local min



Optimization VII

Why do we care then?

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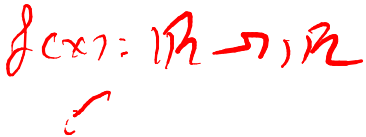
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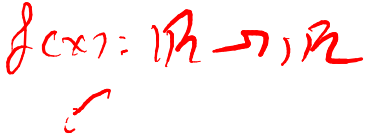
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- Comparing those above values to find the largest (maximum) and smallest (minimum).



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Let's optimize the function $f(x) = \frac{x^5}{5} - \frac{x^4}{4} - \frac{2x^3}{3} + 2$!

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This is where the notion of a sign graph comes in:

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Given a function $f(x)$, the sign graph of $f(x)$ is a useful way to depict only the positivity or negativity of a function.

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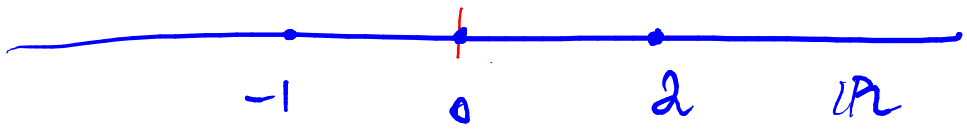
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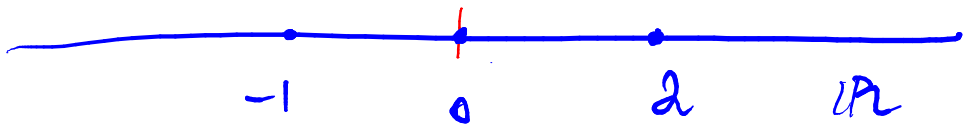
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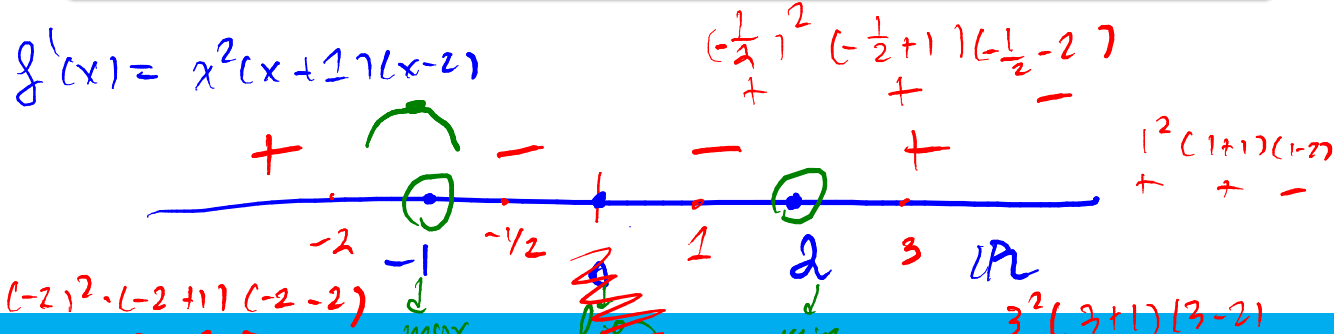


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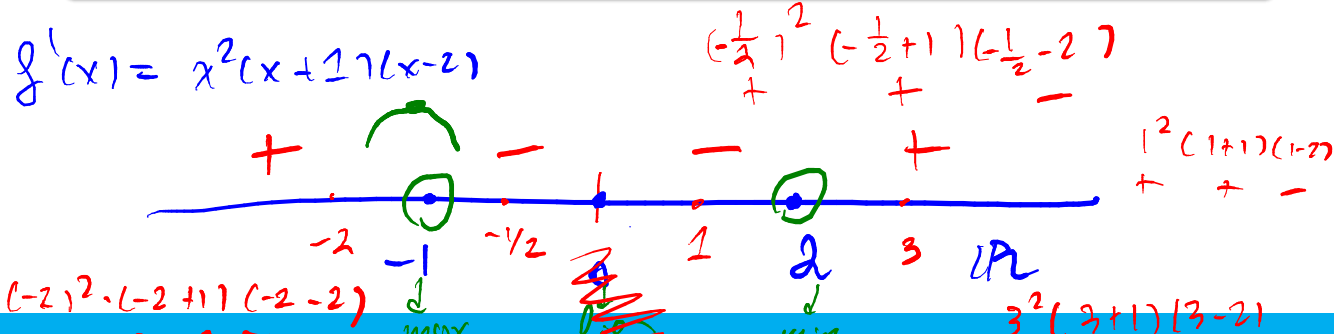


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1) Ami cont. diff ✓

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We start by finding critical points:

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$$f'(x) = 1 + \frac{2}{3\sqrt[3]{x}} \tag{6}$$

$$= \frac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}} \tag{7}$$

Which is zero when:

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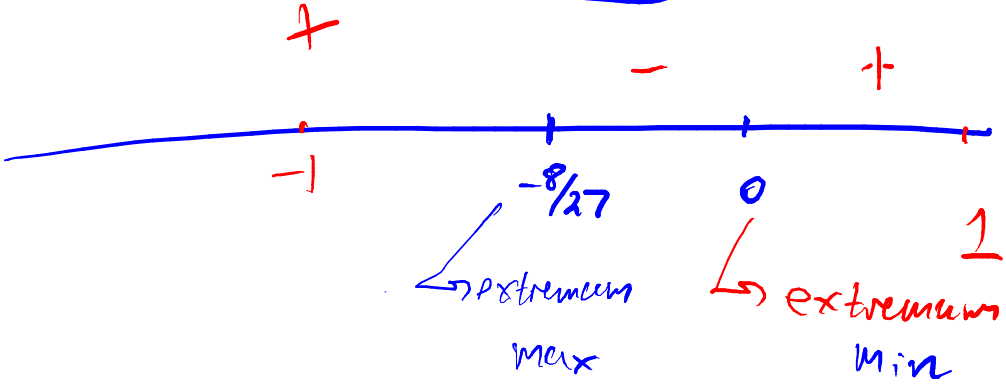
$$\sqrt[3]{x} = 0 \tag{12}$$

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Optimization Example V

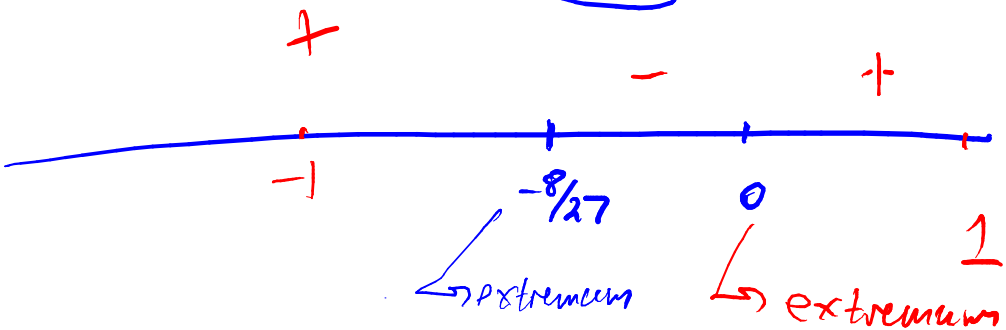
We then construct the sign graph of $f'(x) = \dfrac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}}$.

$\dfrac{2 + 3 \cdot -1}{3 \cdot -1} = \dfrac{-}{-} = +$



Optimization Example V

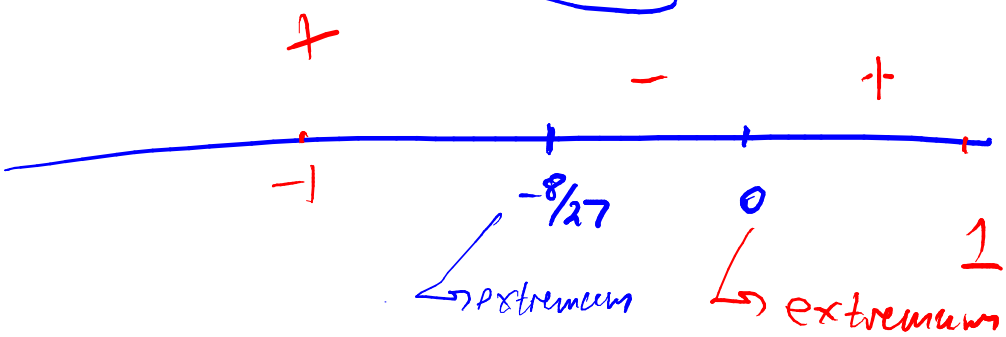
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Which gives us that at $x = -8$ the sign changes from

Optimization Example V

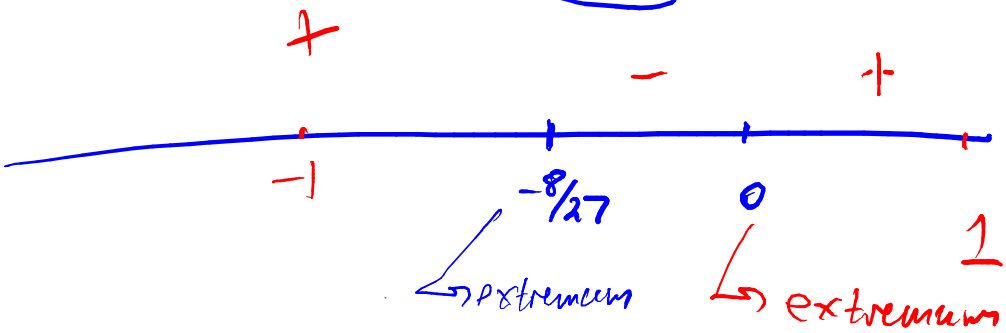
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Optimization Example V

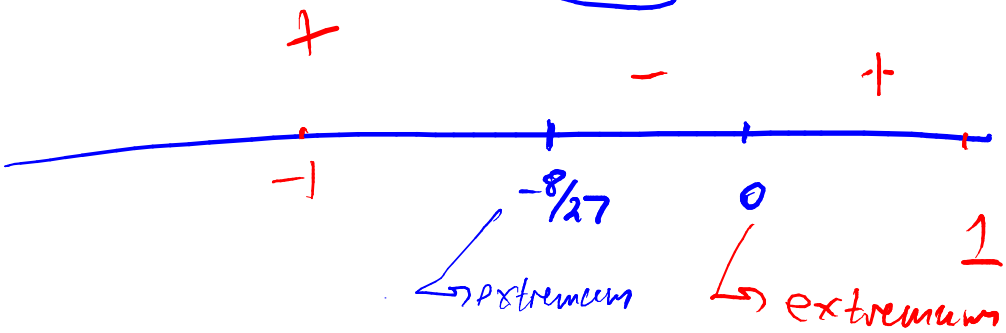
We then construct the sign graph of $f'(x) = \dfrac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}}$.



Which gives us that at $x = -8$ the sign changes from positive to negative and at $x = 0$ the sign changes from

Optimization Example V

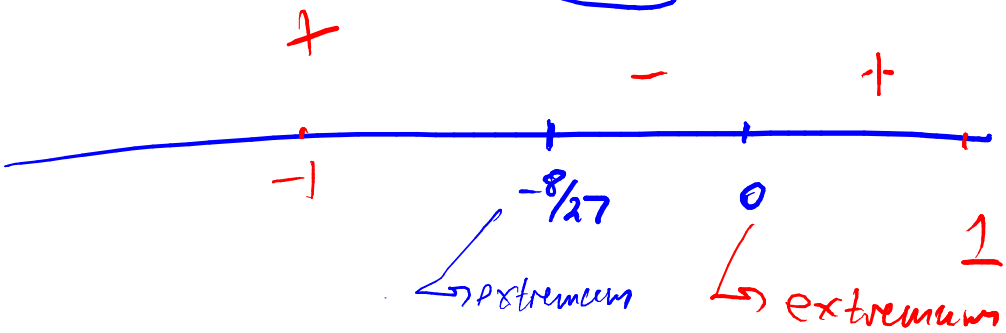
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Optimization Example V

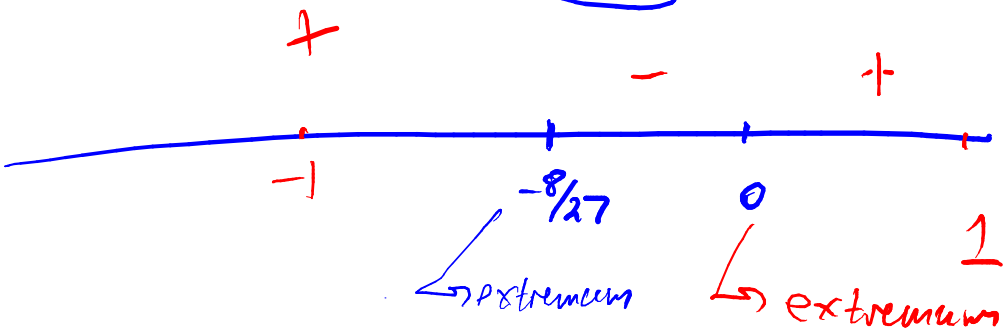
We then construct the sign graph of $f'(x) = \dfrac{2 + 3\sqrt[3]{x}}{3\sqrt[3]{x}}$.



Which gives us that at $x = -8$ the sign changes from positive to negative, and at $x = 0$ the sign changes from negative to positive.

Optimization Example V

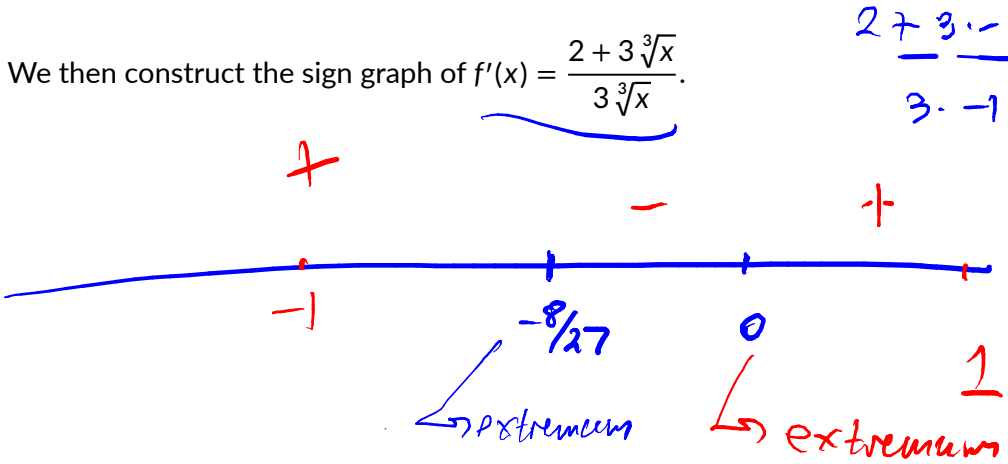
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Which gives us that at $x = -8$ the sign changes from positive to negative, and at $x = 0$ the sign changes from negative to positive.
Therefore $x = -8$ is a local

Optimization Example V

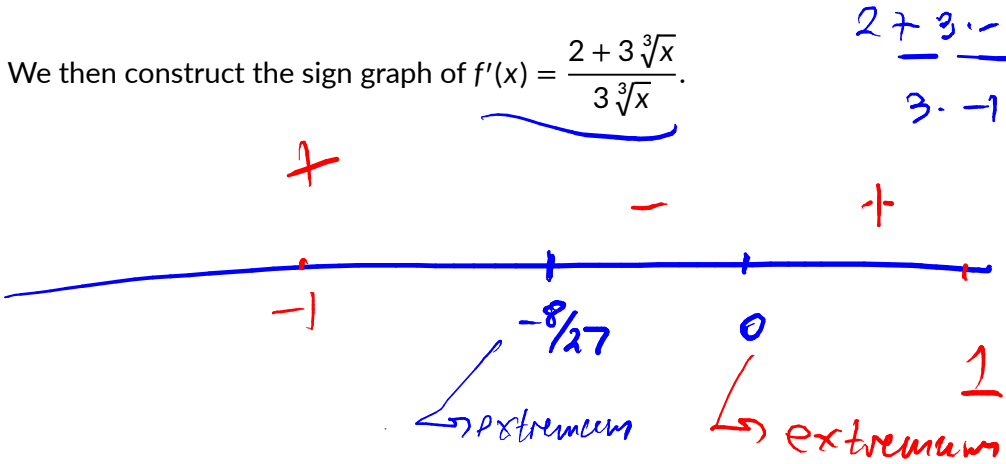
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Optimization Example V

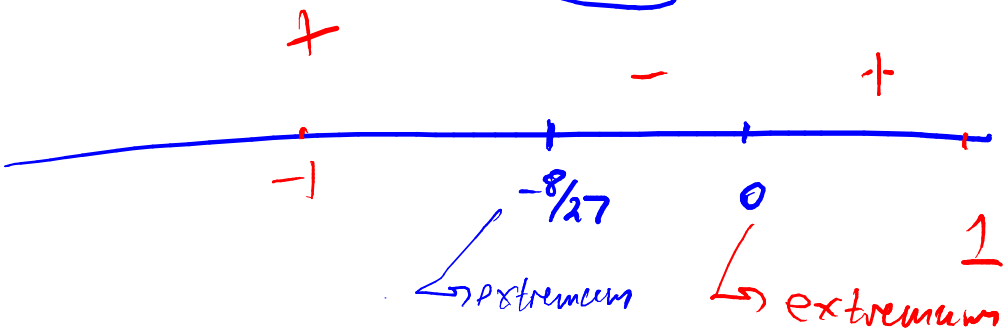
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Optimization Example V

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Which gives us that at $x = -8$ the sign changes from positive to negative, and at $x = 0$ the sign changes from negative to positive.
Therefore $x = -8$ is a local maximum, and $x = 0$ is a local minimum.

Worksheet Time

Let's get into groups and work on the worksheet!

End of Class

That’s all folks!